



Python Programming - 2301CS404

Lab - 6

Roll No. :310

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01) WAP to find sum of all the elements in a List.

```
In [1]: l1 = [1,2,3,4,5,6]
sum=0
for i in l1:
    sum = sum + i
print(sum)
```

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02) WAP to find largest element in a List.

```
In [5]: l1 = [1,76,23,8237,345]
large=l1[0]
for i in l1:
    if(large<i) :
        large = i
print(large)
```

8237

03) WAP to find the length of a List.

```
In [6]: l1 = [23,9,17,635,991]
count=0
for i in l1:
    count+=1
print(count)
```

5

04) WAP to interchange first and last elements in a list.

```
In [2]: n = int(input("Enter the size of the list:"))
l1 = list()
for i in range(n):
    element = int(input("Enter the elements:"))
    l1.append(element)
if(n>1) :
    temp = l1[0]
    l1[0] = l1[-1]
    l1[-1] = temp
print(l1)
```

[5, 2, 3, 4, 1]

05) WAP to split the List into two parts and append the first part to the end.

```
In [9]: n = int(input("Enter the size of the list:"))
l1 = list()
mid_index = n//2
for i in range(n) :
    element = int(input("Enter the elements:"))
    l1.append(element)
for i in range(mid_index):
    temp = l1.pop(0)
    l1.append(temp)
print(l1)
```

[3, 4, 5, 1, 2]

06) WAP to interchange the elements on two positions entered by a user.

```
In [11]: n = int(input("Enter the size of the list:"))
l1 = list()
for i in range(n) :
    element = int(input("Enter the elements:"))
    l1.append(element)
p1 = int(input("Enter the first position:"))
p2 = int(input("Enter the second position:"))
temp = l1[p1]
l1[p1] = l1[p2]
l1[p2] = temp
print(l1)
```

[1, 4, 3, 2, 5]

07) WAP to reverse the list entered by user.

```
In [16]: n = int(input("Enter the size of the list:"))
l1 = list()
l2 = list()
for i in range(n) :
    element = int(input("Enter the elements:"))
    l1.append(element)
l1.reverse()
print(l1)
```

[5, 4, 3, 2, 1]

08) WAP to print even numbers in a list.

```
In [17]: n = int(input("Enter the size of the list:"))
l1 = list()
l2 = list()
for i in range(n) :
    element = int(input("Enter the elements:"))
    l1.append(element)
for i in range(n) :
    if(l1[i]%2==0) :
        l2.append(l1[i])
print(l2)
```

[2, 4]

09) WAP to count unique items in a list.

```
In [18]: n = int(input("Enter the size of the list:"))
l1 = list()
for i in range(n) :
    element = int(input("Enter the elements:"))
    l1.append(element)
l2 = len(set(l1))
print(l2)
```

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10) WAP to copy a list.

```
In [19]: n = int(input("Enter the size of the list:"))
l1 = list()
for i in range(n) :
    element = int(input("Enter the elements:"))
    l1.append(element)
l2 = l1.copy()
print(l2)
```

[2, 4, 6, 8, 10]

11) WAP to print all odd numbers in a given range.

```
In [3]: n = int(input("Enter the size of the list:"))
a = int(input("Enter the starting point:"))
b = int(input("Enter the ending point:"))
l1 = list()
l2 = list()
for i in range(n) :
    element = int(input("Enter the elements:"))
    l1.append(element)
for i in range(a,b+1) :
    if(l1[i]%2!=0) :
        l2.append(l1[i])
print(l2)
```

[1, 3, 5]

12) WAP to count occurrences of an element in a list.

```
In [9]: n = int(input("Enter number of elements: "))
l1 = list()
for i in range(n):
    value = int(input(f"Enter element {i+1}: "))
    l1.append(value)
element = int(input("Enter the element to count: "))
count = l1.count(element)
print(f"The element {element} occurs {count} times in the list.")
```

The element 1 occurs 2 times in the list.

13) WAP to find second largest number in a list.

```
In [12]: n = int(input("Enter number of elements: "))
l1 = list()
for i in range(n):
    element = int(input("Enter the elements:"))
    l1.append(element)
l1.sort()
print(l1[-2])
```

8715

14) WAP to extract elements with frequency greater than K.

```
In [16]: n = int(input("Enter number of elements: "))
l1 = list()
for i in range(n):
    value = int(input(f"Enter element {i+1}: "))
    l1.append(value)
k = int(input("Enter the value of K: "))
l2 = list()
for item in set(l1):
    if l1.count(item) > k:
        l2.append(item)
print("Elements with frequency greater than K:", l2)
```

Elements with frequency greater than K: [1, 2]

15) WAP to create a list of squared numbers from 0 to 9 with and without using List Comprehension.

```
In [17]: n = int(input("Enter the upper limit (e.g., 9): "))
squares = []
for i in range(n + 1):
    squares.append(i * i)
print("List of squared numbers:", squares)
```

List of squared numbers: [0, 1, 4, 9, 16, 25, 36, 49]

16) WAP to create a new list (fruit whose name starts with 'b') from the list of fruits given by user.

```
In [18]: n = int(input("Enter number of fruits: "))
a_fruits = list()
for i in range(n):
    fruit = input(f"Enter fruit {i+1}: ")
    a_fruits.append(fruit)
b_fruits = list()
for fruit in a_fruits:
    if fruit.lower().startswith('b'):
        b_fruits.append(fruit)
print("Fruits starting with 'b':", b_fruits)
```

Fruits starting with 'b': ['banana', 'berry', 'blueberry']

17) WAP to create a list of common elements from given two lists.

```
In [19]: n1 = int(input("Enter number of elements in first list: "))
l1 = list()
for i in range(n1):
    l1.append(int(input(f"Enter element {i+1} of first list: ")))
n2 = int(input("Enter number of elements in second list: "))
l2 = list()
for i in range(n2):
    l2.append(int(input(f"Enter element {i+1} of second list: ")))
common = list()
for item in l1:
    if item in l2 and item not in common:
        common.append(item)
print("Common elements:", common)
```

Common elements: [1, 3, 5]